Assessment Menu: Teacher Guide

The Exponent Bistro · design notes, grading and answer key

Why students cannot just pick the easy ones

1. **Points are capped by category.** Students must complete exactly 4 / 4 / 2 dishes, so they cannot fill the total with Appetizers.
2. **Chef's Specials carry 40% of the grade** and Entrées another 40%. A student who only masters the basics tops out at 20 points.
3. **Skill coverage rule.** Their 10 picks must include at least one dish from each skill group below, so every strand of the unit is assessed.
4. **Harder dishes reward process.** Each has a rubric that gives credit for correct steps, so a partly correct attempt at a harder problem earns more than a perfect easy one in the same time.

Skill group	What it covers	Dishes
RULES	Apply the exponent rules	A1, A5, E1, S1
NEGZERO	Zero and negative exponents	A2, E2, E5, S4
SCINOT	Scientific notation	A3, A6, E3, S2
EXPFUNC	Exponential growth and decay	A4, E4, E6, S3

If a student only does…	Maximum
Only Appetizers (4 × 5)	20
Appetizers + Entrées (4 × 5 + 4 × 10)	60
Full meal (adds 2 Chef's Specials)	100

Grading and options

- Grade only the marked dishes using the rubrics on the next pages (partial credit is built in). Total possible: **100**. If a student marks more than the required number, grade the first in each category.
- Check the skill rule on the order form before totaling. If a skill group is missing, apply your own policy (for example, the dish that broke the rule scores 0, or the student may swap one dish).
- Optional "second helping": allow one extra dish per category that replaces the student's lowest score in that category.
- Look at *which* dishes students chose. Students who select only the easiest dish in each skill group may need a conference.

Student	Appetizers (of 20)	Entrées (of 40)	Specials (of 40)	Skills OK?	Total

Answer Key: Appetizers (5 pts each)

Worked solutions and rubrics. Graphs: draw on the number line; endpoints as noted.

Dish	Problem	Solution	Rubric
A1 RULES	Simplify: $(3x^2)(4x^5)$	**Answer:** $12x^7$ Multiply the coefficients: $3 \times 4 = 12$ Add the exponents: $x^2 \cdot x^5 = x^7$	2 pts: Multiplies coefficients to get 12 2 pts: Adds exponents to get x^7 1 pts: Final answer $12x^7$
A2 NEGZERO	Evaluate each: **(a)** 2^{-3} **(b)** $6^0 + 3^{-2}$	**Answer:** (a) $\frac{1}{8}$ (b) $\frac{10}{9}$ (a) $2^{-3} = \frac{1}{2^3} = \frac{1}{8}$ (b) $6^0 = 1$ and $3^{-2} = \frac{1}{9}$, so the sum is $1 + \frac{1}{9} = \frac{10}{9}$	2 pts: (a) Correct value $\frac{1}{8}$ 1 pts: (b) Uses $6^0 = 1$ 1 pts: (b) Uses $3^{-2} = \frac{1}{9}$ 1 pts: (b) Correct sum $\frac{10}{9}$
A3 SCINOT	Write in scientific notation: **(a)** 52,000 **(b)** 0.0006	**Answer:** (a) 5.2×10^4 (b) 6×10^{-4} (a) Move the decimal 4 places left: 5.2×10^4 (b) Move the decimal 4 places right: 6×10^{-4} (negative exponent for a number less than 1)	1 pts: (a) Coefficient 5.2 1 pts: (a) Exponent 4 1 pts: (b) Coefficient 6 2 pts: (b) Exponent -4 (sign correct)
A4 EXPFUNC	Complete the table for $y = 3 \cdot 2^x$ for x = 0, 1, 2, 3. Is this growth or decay?	**Answer:** y = 3, 6, 12, 24. Growth (the y-values double). $x = 0$: $3 \cdot 2^0 = 3$ $x = 1$: $3 \cdot 2 = 6$; $x = 2$: $3 \cdot 4 = 12$; $x = 3$: $3 \cdot 8 = 24$ The base 2 is greater than 1, so it is exponential **growth**.	1 pts: y = 3 at x = 0 2 pts: y = 6, 12, 24 correct (1 pt for any two) 2 pts: Says growth because the base is greater than 1 (or values double)
A5 RULES	Simplify: **(a)** $\frac{x^8}{x^3}$ **(b)** $(y^3)^4$	**Answer:** (a) x^5 (b) y^{12} (a) Subtract the exponents: $8 - 3 = 5$, so x^5 (b) Multiply the exponents: $3 \times 4 = 12$, so y^{12}	1 pts: (a) Subtracts exponents 2 pts: (a) x^5 2 pts: (b) y^{12} (multiplies exponents)
A6 SCINOT	Which is larger, 3.2×10^5 or 9.1×10^4? Explain how you know.	**Answer:** 3.2×10^5 is larger (320,000 is greater than 91,000). Compare the powers of ten first: 10^5 is ten times 10^4. $3.2 \times 10^5 = 320{,}000$ and $9.1 \times 10^4 = 91{,}000$. So 3.2×10^5 is larger.	2 pts: Correct choice 3.2×10^5 2 pts: Converts to standard form or compares exponents 1 pts: Clear explanation

Answer Key: Entrées (10 pts each)

Worked solutions and rubrics. Graphs: draw on the number line; endpoints as noted.

Dish	Problem	Solution	Rubric
E1 RULES	Simplify, writing with positive exponents: **(a)** $\frac{12x^5y^2}{18x^2y^6}$ **(b)** $(2a^3b^{-2})^3$	**Answer:** (a) $\frac{2x^3}{3y^4}$ (b) $\frac{8a^9}{b^6}$ (a) 12 and 18 reduce to 2 and 3; $x^5 \div x^2 = x^3$; $y^2 \div y^6 = y^{-4} = \frac{1}{y^4}$ (b) $2^3 = 8$, $(a^3)^3 = a^9$, $(b^{-2})^3 = b^{-6} = \frac{1}{b^6}$	2 pts: (a) Reduces the coefficients to 2 and 3 2 pts: (a) x^3 correct 1 pts: (a) y^4 in the denominator 2 pts: (b) 8 and a^9 correct 2 pts: (b) b^{-6} rewritten as b^6 in the denominator 1 pts: Both answers in positive-exponent form
E2 NEGZERO	Simplify with positive exponents: **(a)** $\frac{5x^{-3}}{y^{-2}}$ **(b)** $(3x^{-2})^2$ **(c)** $4m^0 + (4m)^0$	**Answer:** (a) $\frac{5y^2}{x^3}$ (b) $\frac{9}{x^4}$ (c) 5 (a) x^{-3} goes down, y^{-2} goes up: $\frac{5y^2}{x^3}$ (b) $3^2 \cdot x^{-4} = \frac{9}{x^4}$ (c) $4m^0 = 4$ and $(4m)^0 = 1$, so 5	3 pts: (a) Moves both negative exponents correctly 3 pts: (b) Squares 3 and multiplies exponents: $\frac{9}{x^4}$ 1 pts: (c) $4m^0 = 4$ 1 pts: (c) $(4m)^0 = 1$ 2 pts: (c) Correct sum 5 (the zero power applies only to m in the first term)
E3 SCINOT	Compute and write each answer in scientific notation: **(a)** $(3 \times 10^4)(5 \times 10^7)$ **(b)** $\frac{8.4 \times 10^5}{2.1 \times 10^{-2}}$	**Answer:** (a) 1.5×10^{12} (b) 4×10^7 (a) $15 \times 10^{11} =$ **1.5×10^{12}** (b) $8.4 \div 2.1 = 4$ and $10^5 \div 10^{-2} = 10^7$: **4×10^7**	2 pts: (a) Multiplies coefficients and adds exponents (15×10^{11}) 2 pts: (a) Converts to 1.5×10^{12} 3 pts: (b) Divides the coefficients (4) and subtracts exponents correctly 3 pts: (b) Correct answer 4×10^7
E4 EXPFUNC	For x = 0, 1, 2, 3 a table shows y = 80, 120, 180, 270. **(a)** Show the table is exponential. **(b)** Write an equation $y = a \cdot b^x$. **(c)** Find y when x = 4.	**Answer:** (a) common ratio 1.5 (b) $y = 80 \cdot 1.5^x$ (c) y = 405 (a) $\frac{120}{80} = \frac{180}{120} = \frac{270}{180} = 1.5$, a constant ratio. (b) a = 80 (the value at x = 0) and b = 1.5: $y = 80 \cdot 1.5^x$ (c) $270 \times 1.5 = 405$ (or $80 \cdot 1.5^4 = 405$)	3 pts: (a) Divides consecutive y-values to get a constant ratio 3 pts: (b) a = 80 and b = 1.5 2 pts: (b) Equation written correctly 2 pts: (c) y = 405
E5 NEGZERO	Kai wrote: **(a)** $3^{-2} = -9$ **(b)** $6x^0 = 1$ Each answer has a mistake. Explain each mistake and give the correct value.	**Answer:** (a) $\frac{1}{9}$ (b) 6 (a) A negative exponent means a reciprocal, not a negative number: $3^{-2} = \frac{1}{3^2} = \frac{1}{9}$ (b) The zero exponent applies only to x, so $x^0 = 1$ but the 6 stays: $6x^0 = 6$	2 pts: (a) Explains that a negative exponent means reciprocal (not negative) 3 pts: (a) Correct value $\frac{1}{9}$ 2 pts: (b) Explains the exponent 0 applies only to x 3 pts: (b) Correct value 6
E6 EXPFUNC	A $10,000 car loses 20% of its value each year. **(a)** Write an equation for its value V after t years. **(b)** Find its value after 3 years.	**Answer:** (a) $V = 10{,}000(0.8)^t$ (b) $5,120 (c) No: it lost $4,880, which is less than $5,000. (a) Keeps 80% each year: $V = 10{,}000(0.8)^t$	2 pts: (a) Base 0.8 (not 0.2) 1 pts: (a) Initial value 10,000 in the equation 3 pts: (b) Evaluates $10{,}000(0.8)^3 = \$5{,}120$

	(c) Has it lost more than half its value after 3 years? Justify.	(b) 10,000(0.512) = $5,120 (c) Still worth $5,120 (more than $5,000), so it lost only $4,880: **less** than half.	2 pts: (c) Compares with $5,000 2 pts: (c) Correct conclusion: no

Answer Key: Chef’s Specials (20 pts each)

Worked solutions and rubrics. Graphs: draw on the number line; endpoints as noted.

Dish	Problem	Solution	Rubric
S1 RULES	**(a)** Simplify completely, with positive exponents: $\frac{(3x^2y)^3 \cdot 2x^{-1}y^4}{6x^4y^{-2}}$ **(b)** Evaluate your answer when x = 2 and y = 1. **(c)** Name the exponent rules you used.	**Answer:** (a) $9xy^9$ (b) 18 (c) power of a product, product rule, quotient rule (with negative exponents) (a) Numerator: $27x^6y^3 \cdot 2x^{-1}y^4 = 54x^5y^7$. Divide: **$9xy^9$**. (b) 9(2)(1) = 18 (c) Power of a product, product rule, quotient rule.	4 pts: Cubes to $27x^6y^3$ 4 pts: Numerator $54x^5y^7$ 4 pts: Divides: 9, x, y^9 2 pts: (b) 18 4 pts: (c) Names three rules correctly 2 pts: Positive exponents only
S2 SCINOT	Light travels 3×10^8 meters per second. The Sun is 1.5×10^{11} meters from Earth. **(a)** How many seconds does sunlight take to reach Earth? **(b)** A spacecraft moves 3×10^4 meters per second. How long would it take to reach the Sun’s distance? **(c)** How many times longer is (b) than (a)? Use scientific notation.	**Answer:** (a) 5×10^2 s (500 s, about 8.3 min) (b) 5×10^6 s (c) 10^4 times (a) $\frac{1.5 \times 10^{11}}{3 \times 10^8}$ = **5×10^2 s** (b) $\frac{1.5 \times 10^{11}}{3 \times 10^4}$ = **5×10^6 s** (c) $\frac{5 \times 10^6}{5 \times 10^2}$ = **10^4** times longer.	3 pts: (a) Distance ÷ speed 3 pts: (a) 5×10^2 3 pts: (b) Correct setup 3 pts: (b) 5×10^6 4 pts: (c) 10^4 4 pts: Proper scientific notation and units
S3 EXPFUNC	A bacteria culture starts with 500 cells and doubles every 3 hours. Let *h* be the number of 3-hour periods. **(a)** Find the cells after 12 hours. **(b)** What is the first time (in hours) the count exceeds 100,000? **(c)** Another culture has 12,000 cells now and doubles every 3 hours. How many cells were there 9 hours ago?	**Answer:** (a) 8,000 (b) 24 hours (c) 1,500 $N = 500 \cdot 2^h$. (a) h = 4: $500 \cdot 16$ = 8,000. (b) h = 7: 64,000 (too small); h = 8: 128,000. So **24 hours**. (c) 3 doublings back: $\frac{12{,}000}{8}$ = **1,500**.	3 pts: Model $N = 500 \cdot 2^h$ (or equivalent doubling reasoning) 3 pts: (a) Uses h = 4 to get 8,000 6 pts: (b) Tests values (64,000 then 128,000), 3 pts for each check 2 pts: (b) Converts h = 8 to 24 hours 3 pts: (c) Works backward by dividing (not multiplying) by 2 three times 3 pts: (c) Correct answer 1,500
S4 NEGZERO	**(a)** The pattern $2^4 = 16$, $2^3 = 8$, $2^2 = 4$, $2^1 = 2$ continues. Find 2^0, 2^{-1}, 2^{-2} and explain why each makes sense. **(b)** Simplify with positive exponents: $\frac{x^{-3}y^2}{x^{-5}y^{-1}}$ **(c)** Solve: $3^x = \frac{1}{81}$	**Answer:** (a) 1, $\frac{1}{2}$, $\frac{1}{4}$ (b) x^2y^3 (c) x = −4 (a) Each step down divides by 2: $2^0 = 1$, $2^{-1} = \frac{1}{2}$, $2^{-2} = \frac{1}{4}$ (b) $x^{-3+5} = x^2$ and $y^{2+1} = y^3$: **x^2y^3** (c) $\frac{1}{81} = 3^{-4}$, so **x = −4**	2 pts: (a) $2^0 = 1$, $2^{-1} = \frac{1}{2}$, $2^{-2} = \frac{1}{4}$ 4 pts: (a) Explains the pattern: each step divides by 2 3 pts: (b) x^2 (subtracts the negative exponent correctly) 3 pts: (b) y^3 3 pts: (c) Rewrites 81 as 3^4 and the fraction as 3^{-4} 5 pts: (c) Correct answer x = −4